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The AED Model Assessment, Reporting, and Visualisation Library (MARVL) v1.1

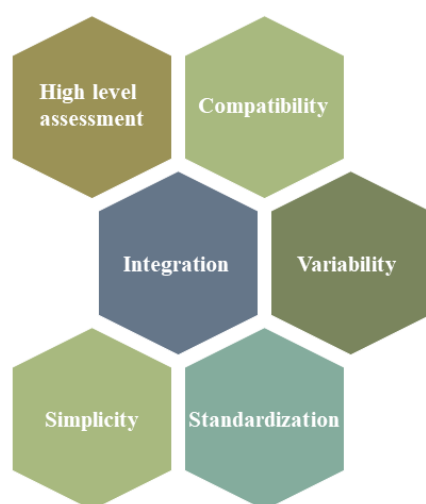
Package Overview



About This Instruction

This document is a User Instruction for the AED Model Assessment, Reporting, and Visualisation Library (MARVL) package. The MARVL package is an integrated software package for visualizing the model outputs and observational datasets, and evaluating the model performance. One of the key challenges for water quality modelling is to assess the model performance against field observations that typically include plenty of biogeochemical variables, exist at multiple sites, and are provided by multiple agencies in different formats. To handle this challenge, the AED research group has developed a series of data processing framework to store the data in a uniform format that is compatible to the AED outputs. At the same time, the group has developed a series of model assessment methods and scripts that were specifically designed to be compatible with the AED data repository and model outputs. The package is a collection of the past AED in-house scripts, and has been refined with uniform and easier user interfaces. This instruction provides guidance for using the MARVL package for visualizing the model outputs and/or data. For the AED variables, units, and science the readers are referred to the AED Science Manual (<https://aquaticecodynamics.github.io/aed-science/>, Hipsey et al., 2022).

MARVL: The AED Model Assessment, Reporting And Visualisation Library



1: High-level assessment

- from state variable assessment to system-scale assessment;

2: Compatibility

- compatible with the 'CSIEM-DATA' data warehouse;
- Compatible with 'TUFLOW-FV' and 'ROMS' outputs

3: Integration

- Integration with 3rd party tools such as tailor diagrams and m_map tools;

4: Variability

- Multiple plotting options to meet high-level assessment needs;

5: Simplicity

- One configuration setting file to integrate model and data;

6: Standardization

- Standard assessment methods and outlooks;



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1. Introduction

1.1 Overview

MARVL is an integrated software package for visualizing the model outputs and observational datasets, and evaluating the model performance. It is also able to carry out high-level system assessment such as environmental response functions and nutrient budgeting analysis, though certain pre-processing steps are required to prepare the data for the high-level system assessment. The current MARVL repository is a collection of AED in-house scripts that have been developed during previous research projects for model assessment, and has been refined with uniform and easier user interfaces. At the moment, the AED-MARVL was designed with the following features:

- Multiple visualizing options such as time-series, transect view, curtain view, site profiling;
- Standard and advanced performance evaluating methods;
- One configuration file for all visualizing functions;
- Uniform outlook (e.g. font style and size, figure resolution) of the figures and animations;
- Option to use YAML style of configurations that can be shared with other plotting software such as R and python;

Following visualizing functions are available in MARVL:

- Time series (with options to show observational data and perform model skill assessment);
- Transect (with options to show observational data on the same transect);
- Transect stacked area for comparing water quality compositions;
- Transect exceedance for comparing water quality levels against guidelines;
- Site profiling of selected water quality variables;
- Sheet plotting and animations;
- Curtain plotting and animations;

High-level systematic assessment of water quality response of to environmental factors and nutrient budgeting analysis have been developed using many of the basic functions of MARVL. However, these high level assessments require several pre-processing steps to prepare data for the assessment, and the configurations and pre-processing varied in different case studies. Therefore, there is still no straightforward script for these high level assessments. We have included the workflow in this document and examples scripts/outputs of the high-level assessments in the MARVL repository. Users of MARVL can follow the workflow and modify the scripts to produce the assessment outcomes of their studies.

The plotting functions have been tested within several case studies, including Coorong Lagoon, Cockburn Sound, and Lake Erie, and examples are included in the current repository. However, MARVL is still in its developing phase. We are expecting feedbacks for us to improve the user experience, as well as functionalities.

1.2 Repository Organisation

The repository is organized as followed:

- ``sources``: plotting scripts and libraries for MARVL;
- ``examples``: place holder for site-specific configurations and recommended place for plotting results;
- ``docs``: place holder for documenting MARVL science and user instructions;

1.3 Execution Instruction

Step 1: Clone the ``aed-marvl`` repository onto local computer

Step 2: Open Matlab (version 2020 or later versions), go to the local ``aed-marvl`` folder and add the paths to tools/libraries by entering

```
addpath(genpath('./'))
```

Step 3: Go to your working folder, edit the `MARVL.m` to configure the plots (use the 'MARVL.m' under other project folder as templates);

Step 4: Under the working folder path, start the plotting by entering

```
run_AEDmarvl('./MARVL.m','matlab')
```

or if you wish to use YAML style configuration

```
run_AEDmarvl('./MARVL.m','yaml')
```

1.4 Update history

- V1.0 – released on 25 Feb 2022, contains the first version of MARVL package (see MARVL user instruction V1.0.docx);
- V1.1 – released on 23 June 2022, with new features of:
 - Add option of Tailor diagram for showing model performance skills;
 - Integrate 'm_map1.4' third-party package for improving sheet animations and plots;
 - Add option of current vectors on to the sheet plotting module;
 - Add option of current vectors on to the transect plotting module;
 - Improve the site profiling module for plotting model outputs at selected sites against mooring records;

2. MARVL Modules

This section introduces the configuration of the MARVL.m file for the visualizing plotting. The configuration file is composed in the form of modules, where the master module is for choosing the plotting functions and defining basic features, and the sub-plotting modules for specifying details in the corresponding plotting functions.

2.1 master module

The configuration of the master module includes choosing the MARVL modules (by commenting/uncommenting out options), model state variables, model path, field data path, and general plotting features. A summary of master module configuration is provided in Table 2.1 and an example is provided after the table.

NOTE: user can include multiple model outputs for time series and transect comparisons.

Configuration Summary

Table 2.1 Summary of master module configuration

Field name	Description	Options	Comments
modules	select plotting function(s) by commenting/uncommenting out the options	'timeseries';...	time-series plot, with option to add observation and performance matrix
		'transect'; ...	transect-view of water quality along a defined longitude
		'transect_stackedArea'; ...	transect-view of water quality components, such as TN vs [PON+DON+NH4+NO3]
		'transect_exceedance'; ...	transect-view of water quality against defined guildlines (thresholds)
		'sheet'; ...	map-view of water quality, with options of 'movie (animation)' and time-averaging
		'curtain'; ...	curtian-view of water quality, with options of 'movie (animation)' and time-averaging
		'site_profiling'; ...	profiling of water quality at selected sites
		'sediment_profiling'; ..	profiling of sediment quality at selected sites (under development)
varname	define water quality variables for plotting	column 1 is the AED variable name (see AED manual for names of the variables), column 2 is the user-defined names in the plot	further options available, including 'TNTP' (calculating TN:TP stoichiometry ratio in time series), 'DINDIP' (DIN:DIP stoichiometry ratio) and 'PONDON' (PON:DON ratio)
ncfile.name	define the path and name of model output file for plotting		user can define multiple model output to be compared in one figure
ncfile.legend	define legend for model to show in plots		user to assign a legend for each model
ncfile.tag	define model type, optional	'TFV' (tuflow) by default, with option 'ROMS' to plot post-processed ROMS output	no tag is required if is using TUFLOW-FV outputs
add_fielddata	option to add field data on timeseries and transect plot	1: add field data; 0: no field data	
fielddata_matfile	path of the field data in .mat format		The field data must be pre-processed according to the AED format for variable names, site descriptions and data organization and saved in .mat format

font	font name for the plots	see font name options in matlab (by entering 'listfonts' in matlab window), 'Times New Roman' by default	
fontsize	Font size in main plot	8 by default	
xlabelsize	font size for x axis label	9 by default	
ylabelsize	font size for y axis label	9 by default	
titlesize	font size for titles	10 by default	
legendsize	font size for legend	6 by default	
visible	option to show figure while plotting	on' or 'off'	

Configuration Example

Example master module configuration:

```
master.modules = {...
    'timeseries';...
    'transect'; ...
    % 'transect_stackedArea'; ...
    % 'transect_exceedance'; ...
    % 'sheet'; ...
    % 'curtain'; ...
    % 'site_profiling'; ...
    % 'sediment_profiling'; ...
};

% state variable Configuration
% column 1: AED names;
% column 2: user-defined names
master.varname = {...
    'SAL','Salinity';...
    'TEMP','Temperature';...
    'WQ_OXY_OXY','DO';...
    'WQ_DIAG_PHY_TCHLA','TCHLA';...
    'WQ_DIAG_TOT_TN','TN';...
    'WQ_DIAG_TOT_TP','TP';...
    'WQ_NIT_AMM','NH_4';...
    'WQ_NIT_NIT','NO_3';...
    'WQ_PHS_FRP','PO_4';...
    'WQ_DIAG_TOT_TSS','TSS';...
    'WQ_DIAG_TOT_TOC','TOC';...
    'WQ_OGM_DOC','DOC';...
    'WQ_DIAG_TOT_TURBIDITY','Turbidity';...
    'DINDIP','DIN/DIP ratio';...
    'PONDON','PON/DON ratio';...
    'TNTP','TN/TP ratio';...
};

master.add_human = 1; % option to use user-define names, if 0 use AED names

% Models
master.ncfile(1).name = 'W:/CDM2022/eWater2022-scenarios-VH-paper-
warmup/output_basecase_005/eWater_basecase_005_all.nc';
master.ncfile(1).legend = 'Basecase';

% field data, stored in standard AED .mat standard format
master.add_fielddata = 1;
master.fielddata_matfile = '.\data\store\archive\cllmm.mat';
master.fielddata = 'cllmm';

% general plotting features
master.font = 'Times New Roman';
master.fontsize = 8;
master.xlabelsize = 9;
master.ylabelsize = 9;
master.titlesize = 10;
master.legendsize = 6;
master.visible = 'off'; % on or off
```


2.2 time-series plotting module

The time-series plotting module is one of the core modules of MARVL. In a basic model performance assessment, the model outputs at the same location and same time of field observations are extracted and the results are compared with the observations to calculate the model skills. However, it is often seen in water quality study that multiple sites (from regular monitoring or occasional surveys) exist in an interested model domain and it is critical for the model to capture the spatial variation rather than just one site. This is especially important for areas with quick and steep bathymetry change, where both the field observations and model present strong spatiotemporal variations. Therefore, this module is designed to be able to show observations from one or multiple sites within a certain area, in comparison to the median and percentiles of model results within the same area. The users can define the size of interested area in GIS polygon files to cover just one site or a larger area with multiple sites.

The module also includes standard and advanced statistic methods to evaluate model performance scores. There are seven statistic methods available in the time-series module for assessing the model performance skills (Table 2.2.1). The readers are referred to these nice publications of Bennett et al. (2013) and Hipsey et al., (2020) for the detailed overview and discussion of these assessment methods. Users can define which method(s) to be included in performance assessment via the configuration of the 'skills' field in the module.

The module provides two options for comparing the observations and model prediction when multiple sites exist at the same time in a define polygon. These options can be configured in the 'scoremethod' field of the module, which include:

- Median: when this option is activated, the median value of all observations is compared to the median value of model predictions. E.g.

$$\begin{cases} O_i = O_{50} \\ P_i = P_{50} \end{cases}$$

where O_{50} and P_{50} are the 50th percentile of observation and model prediction at the time of i th observation, respectively.

- Range: when this option is activated, the median value of all observations is compared to the range of model prediction in the define polygon. For example, if the user define the model range to be 'timeseries.pred_lims = [0.05,0.25,0.5,0.75,0.95]', then the range is taken as the 5th and 95th of model prediction, and the model prediction is considered satisfying if the median observation sit within the range. Otherwise the nearest prediction of the range to the median prediction is used. E.g.

$$\begin{cases} P_i = O_i & \text{if } P_{95} \geq O_i \geq P_5 \\ P_i = P_{95} & \text{if } O_i > P_{95} \\ P_i = P_5 & \text{if } O_i < P_5 \end{cases}$$

where P_5 and P_{95} are the 5th and 95th percentile of model prediction at the time of i th observation, respectively.

Table 2.2.1 Summary of model performance skills in the time series module, where n is the number of observations of O and model prediction of P . O_i and P_i denote the i th observation and prediction, respectively, and $\bar{O}$ and $\bar{P}$ denote the mean value of observations and predictions.

Model skills	Description	Numerical method
R	correlation coefficient	$R = \frac{\sum_{i=1}^N (P_i - \bar{P})(O_i - \bar{O})}{[\sum_{i=1}^N (P_i - \bar{P})^2 \sum_{i=1}^N (O_i - \bar{O})^2]^{1/2}}$
BIAS	bias of model prediction relative to observations	$BIAS = \frac{1}{n} \sum_{i=1}^N (P_i - O_i)$
MAE	mean absolute error	$MAE = \frac{1}{n} \sum_{i=1}^N (P_i - O_i)$
NMAE	normalized mean absolute error to mean observation	$NMAE = MAE / \bar{O}$

RMS	root-mean square error	$RMS = \sqrt{\frac{\sum_{i=1}^N (P_i - O_i)^2}{N}}$
NRMS	normalized root-mean square error to mean observation	$NRMS = RMS/\bar{O}$
MEF	model efficiency, also called Nash-Sutcliffe efficiency	$MEF = 1 - \frac{\sum_{i=1}^N (P_i - O_i)^2}{\sum_{i=1}^N (O_i - \bar{O})^2}$

A summary of time series module configuration is provided in Table 2.2.2. Example configuration and output are provided after the table.

Configuration Summary

Table 2.2.2 Summary of time series module configuration

Field name	Description	Options	Comments
start_plot_ID	select which variable to start plotting		refer to the setting in master.varname
end_plot_ID	select which variable to finish plotting		refer to the setting in master.varname
plotvalidation	option to add field data	1: add field data; 0: no field data	
plotmodel	option to add model results	1: add model results; 0: no model results	
plotdepth	option to select surface or bottom layers	surface and/or bottom	if both surface and bottom are configured, both layers will be added onto same plots and used together for model performance matrix
edge_color	define edge colors for field data symbols		
depth_range	define depth range for plotting		[0 5000] by default
validation_minmax	option to add min and max values of observations	1: add max/min data; 0: no max/min data;	0 by default
isModelRange	option to add model results percentiles	1: add percentile range; 0: no percentile range;	0 by default
pred_lims	define percentile limits for range plotting	5 numbers between 0 -1	[0.05 0.25 0.5 0.75 0.95] by default
alph	transparency of range plot	0 - 1	0.5 by default
isFieldRange	option to add monthly historical field data range	1: add range; 0: no range	0 by default
fielddprctile	define field data percentile range if isFieldRange==1	range between 0 to 100	[10 90] by default
isHTML	option to add all plots into one HTML page	1: add to HTML; 0: no HTML	0 by default
polygon_file	define the polygon file for zones/sites/areas		
plotAllsites	option to plot all sites in the polygon file	1: plot all site; 0: selected sites	1 by default

plotsites	define site IDs for plotting if plotAllsites==0		
add_error	option to calculate model performance matrix	1: calculate; 0: not calculate	0 by default
isSaveErr	option to save the performance matrix in a .mat file	1: save; 0: not save	0 by default
obsTHRESH	define the number of observations over which the performance statistics make sense	>5	5 by default
showSkill	option to show performance statistics in the plot	1: show; 0: not show	0 by default
scoremethod	option to select traditional or advanced method for performance skills	'median' or 'range'	'median': is a traditional method to compare the median observation values to median model results at selected sites; 'range': compare median observation values to the model percentile predictions
skills	option to select statistics skills	R; BIAS; MAE; RMSE; NMAE; NRMS; MEF	
SkillStyle	Option to select performance skill style	'score' or 'tailor'	Score: traditional way to present selected skills in a table on the figure; Tailor: Tailor diagram to present correlation coefficient, RMSE, and standard deviation of both modelled/observed data on the figure.
outputdirectory	define directory to save plots		
htmloutput	define directory to save HTML files		
ErrFilename	define directory and file name to save the performance statistics matrix		
ncfile.symbol	define symbol for model results (surface and bottom)	'-' or '.'	user can define multiple model output to be compared in one figure
ncfile.colour	define colors for model median value plotting (surface and bottom)	RGB format for color definition	
ncfile.col_pal_color_surf	define color for range plot of surface model results		

ncfile.col_pal_color_bot	define color for range plot of bottom model results		
datearray	define time period for plotting		
dateformat	define time format to show in x axis	mm/yy' by default	
istitled	option to add title	1: add; 0: not add	1 by default
isylabel	option to add y label	1: add; 0: not add	1 by default
islegend	option to add legend	1: add; 0: not add	1 by default
isYlim	option to define Y axis limits	1: add; 0: not add	0 by default
isGridon	option to add grid on	1: add; 0: not add	1 by default
dimensions	define figure dimensions in centimeters	[20 10] by default	
dailyave	option to use daily average or raw model output internal	1: daily-average; 0: raw model output interval	0 by default
smoothfactor	option to smooth out the median model results	use odd numbers, such as 1 or 3	1 by default, no smoothing
legendlocation	define legend location	'northeastoutside' by default	
filetype	define figure file type	'png' or 'eps'	'png' to save figures to PNG format only; 'eps' to save figures in both EPS and 300dpi JPG formats
cAxis.value	define limits of Y axis	empty [] by default, matlab will automatically adjust the y limit	

Configuration Example

Example timeseries module configuration:

```
timeseries.start_plot_ID = 1; % select which variable to plot
timeseries.end_plot_ID = 12;

timeseries.plotvalidation = 1; % Add field data to figure (1 or 0)
timeseries.plotmodel = 1;

timeseries.plotdepth = {'surface','bottom'}; % {'surface','bottom'} Cell-array with either one
timeseries.edge_color = {[166,86,40]./255;[8,88,158]./255]; % symbol edge color for field data, surface and bottom
timeseries.depth_range = [0.2 100];
timeseries.validation_minmax = 0; % option to add max/min observations
timeseries.isModelRange = 1; % option to plot model range with below percentile
timeseries.pred_lims = [0.05,0.25,0.5,0.75,0.95]; % must be 5 numbers
timeseries.alph = 0.5; % transparency

timeseries.isFieldRange = 0; % option to add plot field data range
timeseries.fieldprctile = [10 90];
timeseries.isHTML = 1;

% polygon file define the site areas
timeseries.polygon_file = 'E:\database\AED-MARVL-v0.4\Projects\Erie\GIS\erie_validation_v4.shp';

% option to plot all sites or selected sites
timeseries.plotAllsites = 1;
if timeseries.plotAllsites == 0
    timeseries.plotsite = [1];
end

% section for model skill calculations
timeseries.add_error = 1;
timeseries.isSaveErr = 1;
timeseries.obsTHRESH = 5;
timeseries.showSkill = 1;
timeseries.scoremethod = 'range'; % 'range' or 'median'
timeseries.SkillStyle = 'score'; % 'score' for score table or 'tailor' for tailor diagram

% selection of model skill assessment, 1: activated; 0: not activated
timeseries.skills = [1,... % r: regression coefficient (0-1)
    1,... % BIAS: bias relative to mean observation (%)
    1,... % MAE: mean absolute error
    1,... % RMSE: root mean square error
    1,... % NMAE: MAE normalized to mean observation
    1,... % NRMS: RMSE normalized to mean observation
    1,... % MEF: model efficiency, Nash-Sutcliffe Efficiency
    ];
timeseries.outputdirectory = 'E:\database\AED-MARVL-v0.4\Projects\Erie\plotting\timeseries_testF\RAW';
timeseries.htmloutput = 'E:\database\AED-MARVL-v0.4\Projects\Erie\plotting\timeseries_testF\HTML';
timeseries.ErrFilename = 'E:\database\AED-MARVL-v0.4\Projects\Erie\plotting\timeseries_testF\errormatrix.mat';

timeseries.ncfile(1).symbol = {'-','-'};
timeseries.ncfile(1).colour = {[166,86,40]./255;[8,88,158]./255]; % Surface and Bottom
timeseries.ncfile(1).col_pal_color_surf = [[254,232,200]./255;[252,141,89]./255]; % color1: 5-95 perc; color2: 25-75 perc
timeseries.ncfile(1).col_pal_color_bot = [[222,235,247]./255;[107,174,214]./255];

% plotting configuration
```

```

timeseries.datearray = datenum(2013,5:1:10,01);
timeseries.dateformat = 'mm/yy';

%timeseries.dimc = [0.9 0.9 0.9]; % dimmest (lightest) color
timeseries.istitled = 1;
timeseries.isylabel = 1;
timeseries.islegend = 1;
timeseries.isYlim = 1;
%timeseries.isGridon = 1;
timeseries.dimensions = [15 7.5]; % Width & Height in cm

timeseries.dailyave = 0; % 1 for daily average, 0 for model output interval.

timeseries.legendlocation = 'northeastoutside';
timeseries.filetype = 'eps';

for vvv=1:size(MARVLs.master.varname,1)
    timeseries.cAxis(vvv).value = [ ];
end

```

Example Output

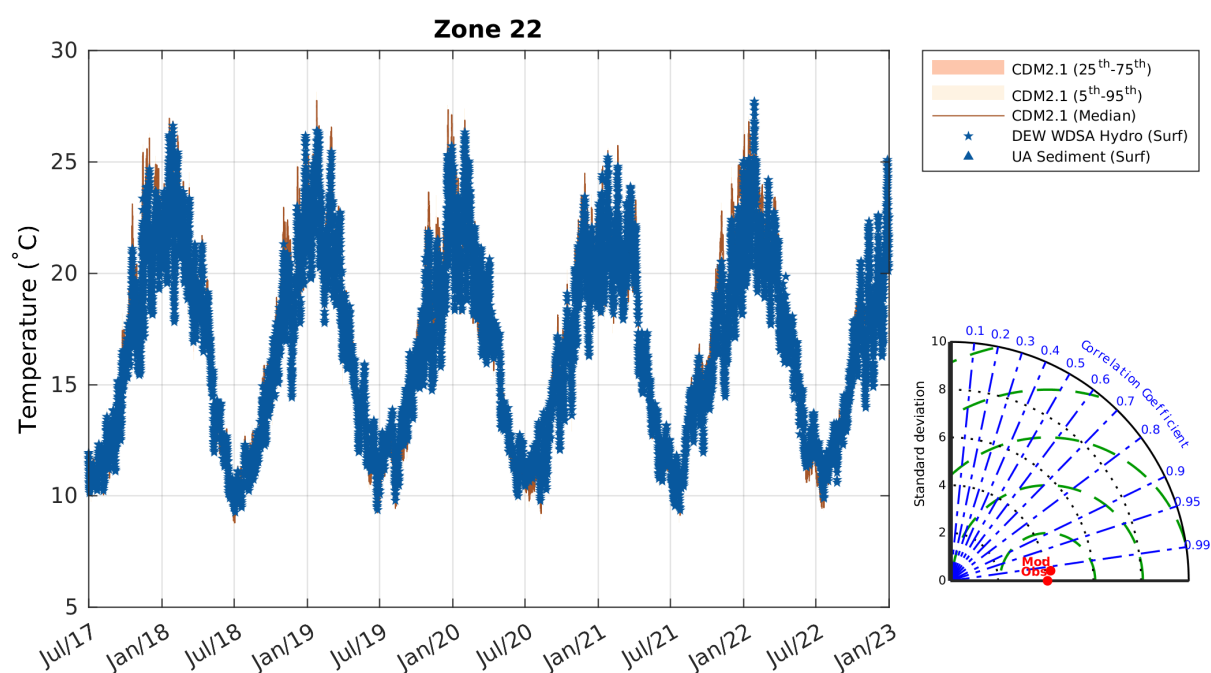


Figure 2.2. Example output of time-series module for water temperature in Coorong South Lagoon with the 'Range' method and Taylor diagram for model performance assessment.

2.3 transect plotting module

The transect plotting module is used for producing model predictions along a defined polyline, and has options to include observations within certain distance to the polyline. This module is useful especially in study site with significant spatial variations, such as an estuary crossing a freshwater-marine spectrum, or coastal system with shallow to deep water gradients. Users can define time period(s) for the transect, add model prediction ranges and box plots of observations on top of model results. A summary of transect module configuration is provided in Table 2.3. Example configuration and output are provided after the table.

Configuration Summary

Table 2.3 Summary of transect module configuration

Field name	Description	Options	Comments
start_plot_ID	select which variable to start plotting		refer to the setting in master.varname
end_plot_ID	select which variable to finish plotting		refer to the setting in master.varname
plotvalidation	option to add field data	1: add field data; 0: no field data	
plotmodel	option to add model results	1: add model results; 0: no model results	
isRange	option to add model results percentiles	1: add percentile range; 0: no percentile range;	0 by default
pred_lims	define percentile limits for range plotting	5 numbers between 0 -1	[0.05 0.25 0.5 0.75 0.95] by default
alph	transparency of range plot	0 - 1	0.5 by default
isFieldRange	option to add monthly historical field data range	1: add range; 0: no range	0 by default
isHTML	option to add all plots into one HTML page	1: add to HTML; 0: no HTML	0 by default
polygon_file	define the line GIS file for transect		
outputdirectory	define directory to save plots		
htmloutput	define directory to save HTML files		
ncfile.symbol	define symbol for model results	'-' or '.'	user can define multiple model output to be compared in one figure
ncfile.colour	define colors for model median value plotting	RGB format for color definition	
ncfile.col_pal_color	define color for range plot of surface model results		
istitled	option to add title	1: add; 0: not add	1 by default
isylabel	option to add y label	1: add; 0: not add	1 by default
islegend	option to add legend	1: add; 0: not add	1 by default
isYlim	option to define Y axis limits	1: add; 0: not add	0 by default
isGridon	option to add grid on	1: add; 0: not add	1 by default

isSurf	option to choose surface or bottom layer	1: surface layer; 0: bottom layer	1 by default
isSpherical	option to define the coordinate type	1: spherical; 0: Euler	0 by default
add_obs-num	option to add number of observations	1: add; 0: not add	0 by default
boxlegend	define field data box plot legend location		'southeast' by default
rangelegend	define model result legend location		'northwest' by default
dimensions	define figure dimensions in centimeters	[20 10] by default	
pdates.value	time period for transect plots		can define multiple periods
binfielddata	option to include nearby field data	1: include; 0: not include	1 by default
binradius	define the radius of bins for field data search	number with unit of km	0.5 by default
linedist	distance from model polyline to be considered	number with unit of m	500 by default
xlim	define limits in X axis		
xticks	define ticks in X axis		
xlabel	define label on X axis		
cAxis.value	define limits of Y axis	empty [] by default, matlab will automatically adjust the y limit	
filetype	define figure file type	'png' or 'eps'	'png' to save figures to PNG format only; 'eps' to save figures in both EPS and 300dpi JPG formats

Configuration Example

Example transect plotting module configuration

```
transect.start_plot_ID = 1;
transect.end_plot_ID = 1;

transect.polygon_file = 'E:\database\AED-MARVL-0.1\GIS\Coorong\Transect_Coorong.shp';
% Add field data to figure
transect.plotvalidation = 1; % 1 or 0
transect.plotmodel = 1; % 1 or 0
transect.isRange = 1;
transect.pred_lims = [0.05,0.25,0.5,0.75,0.95];

transect.istitled = 1;
transect.isylabel = 1;
transect.islegend = 1;
transect.isYlim = 1;
transect.isHTML = 1;
transect.isSurf = 1; %plot surface (1) or bottom (0)
transect.isSpherical = 0;
%transect.use_matfiles = 0;
transect.add_obs_num = 1;
%config.boxon = 1;

% ____
transect.outputdirectory = 'E:\database\AED-MARVL-v0.4\Projects\Coorong\testF\RAW/';
transect.htmloutput = 'E:\database\AED-MARVL-v0.4\Projects\Coorong\testF\HTML/';

% plotting configuration
%transect.dimc = [0.9 0.9 0.9]; % dimmest (lightest) color
transect.boxlegend = 'southeast';
transect.rangelegend = 'northwest';
transect.dimensions = [20 10]; % Width & Height in cm
transect.filetype = 'eps';

for i=1:20
transect.pdates(i).value = [datenum(2017,7+(i-1)*3,01) datenum(2017,07+i*3,01)-1];
end
transect.binfielddata = 1;
% radius distance to include field data. Used to bin data where number of
% sites is higher, but the frequency of sampling is low. The specified
% value will also make where on the line each polygon will be created. So
% if radius == 5, then there will be a search polygon found at r*2, so 0km, 10km, 20km etc. In windy rivers these polygons may
% overlap.
transect.binradius = 0.5;% in km;

%distance from model polyline to be considered.
%Field data further than specified distance won't be included.
%Even if found with search radius. This is to attempt to exclude data
%sampled outside of the domain.
transect.linedist = 1500;% in m

transect.xlim = [0 110];% xlim in KM
transect.xticks = [0:10:110];
transect.xlabel = 'Distance from Goolwa to South Lagoon (km)';

transect.cAxis(1).value = [0 200];
```

```

% ylim
for vvvv=2:size(MARVLs.master.varname,1)
    transect.cAxis(vvvv).value = [ ];
end
transect.ncfile(1).symbol = {'-'};
transect.ncfile(1).translate = 1;
transect.ncfile(1).colour = [166,86,40]./255;% Surface and Bottom
transect.ncfile(1).edge_color = [166,86,40]./255;
transect.ncfile(1).col_pal_color = [[176 190 197]./255;[162 190 197]./255;[150 190 197]./255;[150 190 197]./255];

transect.ncfile(2).symbol = {'-'};
transect.ncfile(2).translate = 1;
transect.ncfile(2).colour = [27,158,119]./255;% Surface and Bottom
transect.ncfile(2).edge_color = [8,88,158]./255;
transect.ncfile(2).col_pal_color = [[102,194,164]./255;[162 190 197]./255;[150 190 197]./255;[150 190 197]./255];

```

Example Output

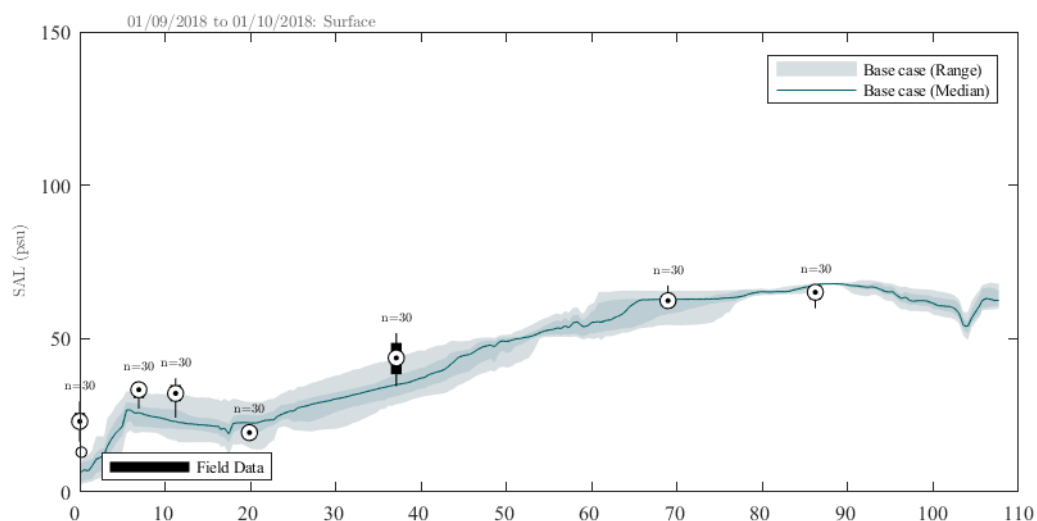


Figure 2.3. Example output of transect module for salinity in the Coorong in Sep 2018 in comparison to the observations.

2.4 transect stacked area plotting module

The transect stacked area plotting module is a variant of the transect plotting module, and able to perform comparison of model predicted water quality components to the total observed concentration along a transect, such as comparing the modelled biomass of phytoplankton groups to the observed total chlorophyll-a concentrations. A summary of transect stacked area module configuration is provided in Table 2.4. Example configuration and output are provided after the table.

Configuration Summary

Table 2.4 Summary of transect stacked area module configuration

Field name	Description	Options	Comments
thevars	variable names of components		refer to the setting in master.varname
thelabels	labels of variables of components		refer to the setting in master.varname
thevars_conv	conversion factor of components		for example, 12/50 to convert phytoplankton biomass of mmol C/m ² to ug/L
isHTML	option to add all plots into one HTML page	1: add to HTML; 0: no HTML	0 by default
polygon_file	define the line GIS file for transect		
outputdirectory	define directory to save plots		
htmloutput	define directory to save HTML files		
istitled	option to add title	1: add; 0: not add	1 by default
isylabel	option to add y label	1: add; 0: not add	1 by default
islegend	option to add legend	1: add; 0: not add	1 by default
isYlim	option to define Y axis limits	1: add; 0: not add	0 by default
isGridon	option to add grid on	1: add; 0: not add	1 by default
isSurf	option to choose surface or bottom layer	1: surface layer; 0: bottom layer	1 by default
isSpherical	option to define the coordinate type	1: spherical; 0: Euler	0 by default
add_obs-num	option to add number of observations	1: add; 0: not add	0 by default
boxlegend	define field data box plot legend location		'southeast' by default
rangelegend	define model result legend location		'northwest' by default
dimensions	define figure dimensions in centimeters	[20 10] by default	
pdates.value	time period for transect plots		can define multiple periods
binfielddata	option to include nearby field data	1: include; 0: not include	1 by default
binradius	define the radius of bins for field data search	number with unit of km	0.5 by default

linedist	distance from model polyline to be considered	number with unit of m	500 by default
xlim	define limits in X axis		
xticks	define ticks in X axis		
xlabel	define label on X axis		
cAxis.value	define limits of Y axis	empty [] by default, matlab will automatically adjust the y limit	
filetype	define figure file type	'png' or 'eps'	'png' to save figures to PNG format only; 'eps' to save figures in both EPS and 300dpi JPG formats

Configuration Example

Example transect stacked area configuration

```
transectSA.thevars = {...
    'WQ_PHY_GRN';...
    'WQ_PHY_DINO';...
    'WQ_PHY_DIATOM';...
    'WQ_PHY_CRYPT';...
};
transectSA.thelabels = {...
    'Green Algae';...
    'Dino';...
    'Diatom';...
    'Crypt';...
};
transectSA.thevars_conv = [50 50 50 50]/12;

transectSA.varname = {...
    'WQ_DIAG_PHY_TCHLA','TChla (\mug/L)';...
};
%transectSA.add_human = 1;

transectSA.polygon_file = 'E:\database\AED-MARVI-v0.2\Examples\Cockburn\GIS\Curtain_polyline_100m_QC.shp';
% Add field data to figure
transectSA.plotvalidation = 1; % 1 or 0
transectSA.pred_lims = [0.05,0.25,0.5,0.75,0.95];

transectSA.isRange = 1;
transectSA.istitled = 1;
transectSA.isylabel = 1;
transectSA.islegend = 0;
transectSA.isYlim = 1;
transectSA.isHTML = 1;
transectSA.isSurf = 1; %plot surface (1) or bottom (0)
transectSA.isSpherical = 0;
transectSA.use_matfiles = 0;
transectSA.add_obs_num = 1;
%config.boxon = 1;

% ____
transectSA.outputdirectory = 'plotting/transectSA/RAW/';
transectSA.htmloutput = 'plotting/transectSA/HTML/';

% plotting configuration
transectSA.dimc = [0.9 0.9 0.9]; % dimmest (lightest) color
transectSA.boxlegend = 'southeast';
transectSA.rangelegend = 'northwest';
transectSA.dimensions = [20 10]; % Width & Height in cm

i=1;
transectSA.pdates(1).value = [datetime(2021,06,16) datetime(2021,07,01)];i=i+1;
transectSA.pdates(2).value = [datetime(2021,07,01) datetime(2021,07,15)];i=i+1;
transectSA.pdates(3).value = [datetime(2021,07,16) datetime(2021,08,01)];i=i+1;

transectSA.binfielddata = 1;
% radius distance to include field data. Used to bin data where number of
% sites is higher, but the frequency of sampling is low. The specified
% value will also make where on the line each polygon will be created. So
```

% if radius == 5, then there will be a search polygon found at $r*2$, so 0km, 10km, 20km etc. In windy rivers these polygons may overlap.

transectSA.binradius = 0.5;% in km;

%distance from model polyline to be considered.

%Field data further than specified distance won't be included.

%Even if found with search radius. This is to attempt to exclude data

%sampled outside of the domain.

transectSA.linedist = 1500;% in m

transectSA.xlim = [0 45];% xlim in KM

transectSA.xticks = [0:5:45];

transectSA.xlabel = 'Distance from Southern CS (km)';

% ylim

transectSA.cAxis(1).value = [0 20];

Example Output

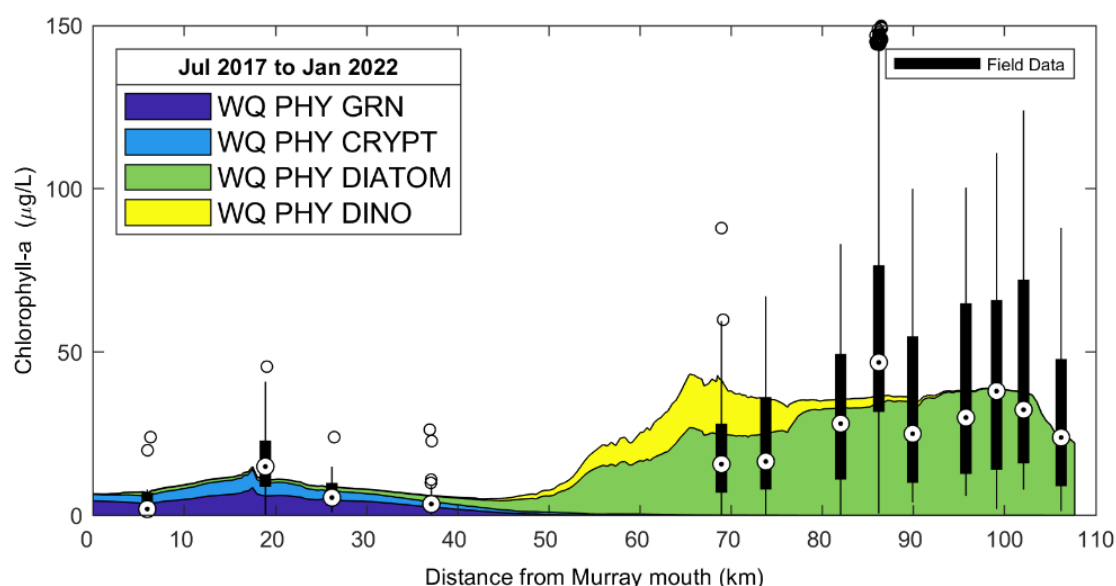


Figure 2.3. Example output of transect stacked area plot for phytoplankton biomass in the Coorong from 07/2017 to 01/2022.

2.5 transect exceedance plotting module

The transect exceedance plotting module is also a variant of the transect plotting module, and able to perform comparison of model predicted water quality against environmental guidelines or thresholds, such as comparing the predicted water quality concentrations to the ANZECC guidelines. A summary of transect exceedance module configuration is provided in Table 2.5. Example configuration and output are provided after the table.

Configuration Summary

Table 2.5 Summary of transect exceedance module configuration

Field name	Description	Options	Comments
start_plot_ID	select which variable to start plotting		refer to the setting in master.varname
end_plot_ID	select which variable to finish plotting		refer to the setting in master.varname
plotvalidation	option to add field data	1: add field data; 0: no field data	only model data to be plotted in this module, need to be 0
plotmodel	option to add model results	1: add model results; 0: no model results	only model data to be plotted in this module, need to be 1
isHTML	option to add all plots into one HTML page	1: add to HTML; 0: no HTML	0 by default
polygon_file	define the line GIS file for transect		
outputdirectory	define directory to save plots		
htmloutput	define directory to save HTML files		
istitled	option to add title	1: add; 0: not add	1 by default
isylabel	option to add y label	1: add; 0: not add	1 by default
islegend	option to add legend	1: add; 0: not add	1 by default
isYlim	option to define Y axis limits	1: add; 0: not add	0 by default
isGridon	option to add grid on	1: add; 0: not add	1 by default
isSurf	option to choose surface or bottom layer	1: surface layer; 0: bottom layer	1 by default
isSpherical	option to define the coordinate type	1: spherical; 0: Euler	0 by default
thresh.value	threshold values to be added onto plots		user can define multiple thresholds
thresh.legend	legends for thresholds		string array match the value size
boxlegend	define field data box plot legend location		'southeast' by default
rangelegend	define model result legend location		'northwest' by default
dimensions	define figure dimensions in centimeters	[20 10] by default	

pdates.value	time period for transect plots		can define multiple periods
binfielddata	option to include nearby field data	1: include; 0: not include	1 by default
binradius	define the radius of bins for field data search	number with unit of km	0.5 by default
linedist	distance from model polyline to be considered	number with unit of m	500 by default
xlim	define limits in X axis		
xticks	define ticks in X axis		
xlabel	define label on X axis		
cAxis.value	define limits of Y axis	empty [] by default, matlab will automatically adjust the y limit	
filetype	define figure file type	'png' or 'eps'	'png' to save figures to PNG format only; 'eps' to save figures in both EPS and 300dpi JPG formats

Configuration Example

Example transect exceedance module configuration

```
transectExc.start_plot_ID = 13;
transectExc.end_plot_ID = 13;

transectExc.polygon_file = 'E:\database\AED-MARVI-v0.2\Examples\Cockburn\GIS\Curtain_polyline_100m_QC.shp';
% Add field data to figure
transectExc.plotvalidation = 0; % 1 or 0
transectExc.pred_lims = [0.05,0.25,0.5,0.75,0.95];

transectExc.isRange = 1;
transectExc.istitled = 1;
transectExc.isylabel = 1;
transectExc.islegend = 0;
transectExc.isYlim = 1;
transectExc.isHTML = 1;
transectExc.isSurf = 1; %plot surface (1) or bottom (0)
transectExc.isSpherical = 0;
%transectExc.use_matfiles = 0;
%transectExc.add_obs_num = 1;
%config.boxon = 1;

transectExc.thresh(13).value = [0.5 1];
transectExc.thresh(13).legend = {'%time > 0.5NTU',...
    '%time > 1NTU'};
% ____
transectExc.outputdirectory = 'plotting/transect_exceedance/RAW/';
transectExc.htmloutput = 'plotting/transect_exceedance/HTML/';

% plotting configuration
transectExc.dimc = [0.9 0.9 0.9]; % dimmest (lightest) color
transectExc.boxlegend = 'southeast';
transectExc.rangelegend = 'northwest';
transectExc.dimensions = [20 10]; % Width & Height in cm

i=1;
transectExc.pdates(1).value = [datenum(2021,06,16) datenum(2021,07,01)];i=i+1;
transectExc.pdates(2).value = [datenum(2021,07,01) datenum(2021,07,15)];i=i+1;
transectExc.pdates(3).value = [datenum(2021,07,16) datenum(2021,08,01)];i=i+1;

transectExc.binfielddata = 1;
% radius distance to include field data. Used to bin data where number of
% sites is higher, but the frequency of sampling is low. The specified
% value will also make where on the line each polygon will be created. So
% if radius == 5, then there will be a search polygon found at r*2, so 0km, 10km, 20km etc. In windy rivers these polygons may
% overlap.
transectExc.binradius = 0.5;% in km;

%distance from model polyline to be considered.
%Field data further than specified distance won't be included.
%Even if found with search radius. This is to attempt to exclude data
%sampled outside of the domain.
transectExc.linedist = 1500;% in m

transectExc.xlim = [0 45];% xlim in KM
transectExc.xticks = [0:5:45];
```

```

transectExc.xlabel = 'Distance from Southern CS (km)';

% ylim
for vvv=1:size(MARVLs.master.varname,1)
    transectExc.cAxis(vvv).value = [0 100];
end

transectExc.ncfile(1).symbol = {'-'};
transectExc.ncfile(1).translate = 1;
transectExc.ncfile(1).colour = [166,86,40]./255;% Surface and Bottom
transectExc.ncfile(1).edge_color = [166,86,40]./255;
transectExc.ncfile(1).col_pal_color = [[176 190 197]./255;[162 190 197]./255;[150 190 197]./255;[150 190 197]./255];

```

Example Output

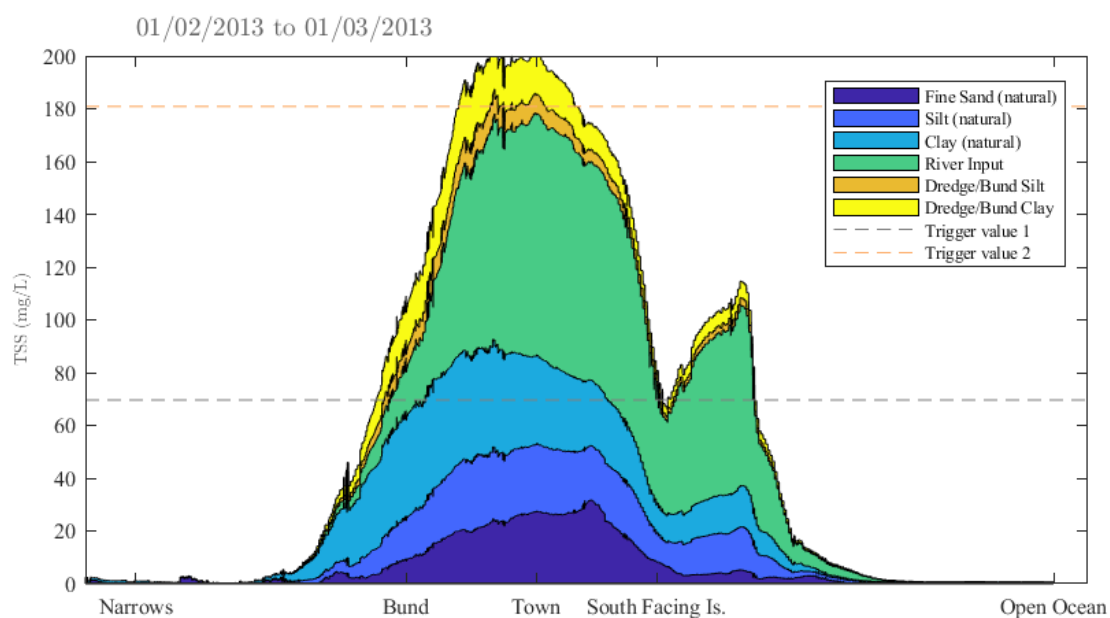


Figure 2.5. Example transect exceedance plot for suspended solids concentration against guidelines in Gladstone.

2.6 site profiling plotting module

The site profiling plotting module is a simple module for plotting selected water quality at one or multiple sites. A summary of site profiling module configuration is provided in Table 2.6. Example configuration and output are provided after the table.

Configuration Summary

Table 2.5 Summary of site profiling module configuration

Field name	Description	Options	Comments
start_plot_ID	select which variable to start plotting		refer to the setting in master.varname
end_plot_ID	select which variable to finish plotting		refer to the setting in master.varname
sitenames	names of plotting sites		can define multiple sites
siteX	X coordinates of sites		same array size to site names
siteY	Y coordinates of sites		same array size to site names
plotvalidation	option to add field data	1: add field data; 0: no field data	currently only model data can be plotted
plotmodel	option to add model results	1: add model results; 0: no model results	
isHTML	option to add all plots into one HTML page	1: add to HTML; 0: no HTML	0 by default
outputdirectory	define directory to save plots		
htmloutput	define directory to save HTML files		
datearray	define time period for plotting		
dateformat	define time format to show in x axis	mm/yy' by default	
istitled	option to add title	1: add; 0: not add	1 by default
isylabel	option to add y label	1: add; 0: not add	1 by default
islegend	option to add legend	1: add; 0: not add	1 by default
isYlim	option to define Y axis limits	1: add; 0: not add	0 by default
isGridon	option to add grid on	1: add; 0: not add	1 by default
dimensions	define figure dimensions in centimeters	[20 10] by default	
filetype	define figure file type	'png' or 'eps'	'png' to save figures to PNG format only; 'eps' to save figures in both EPS and 300dpi JPG formats
cAxis.value	define limits of Y axis	empty [] by default, matlab will automatically adjust the y limit	

Configuration Example

Example site profiling module configuration

```
profile.start_plot_ID = 1;
profile.end_plot_ID = 1;

profile.sitenames={'Cockburn','Swan'};
profile.siteX=[ 380000, 388340];
profile.siteY=[6433760,6458300];

profile.plotvalidation = false; % Add field data to figure (true or false)
profile.plotmodel = 1;

profile.filetype = 'eps';
profile.expected = 1; % plot expected WL

profile.isHTML = 1;

profile.datearray = datenum(2021,6,15:15:60);

profile.dateformat = 'dd/mm/yyyy';
for vvv=1:length(MARVLs.master.varname)
    profile.cAxis(vvv).value = [ ];
end

profile.dimc = [0.9 0.9 0.9]; % dimmest (lightest) color
profile.istitled = 1;
profile.isylabel = 1;
profile.islegend = 1;
profile.isYlim = 1;
profile.isGridon = 1;
profile.dimensions = [20 10]; % Width & Height in cm

profile.outputdirectory = 'plotting/profile/RAW/';
profile.htmloutput = 'plotting/profile/HTML/';
```

Example Output

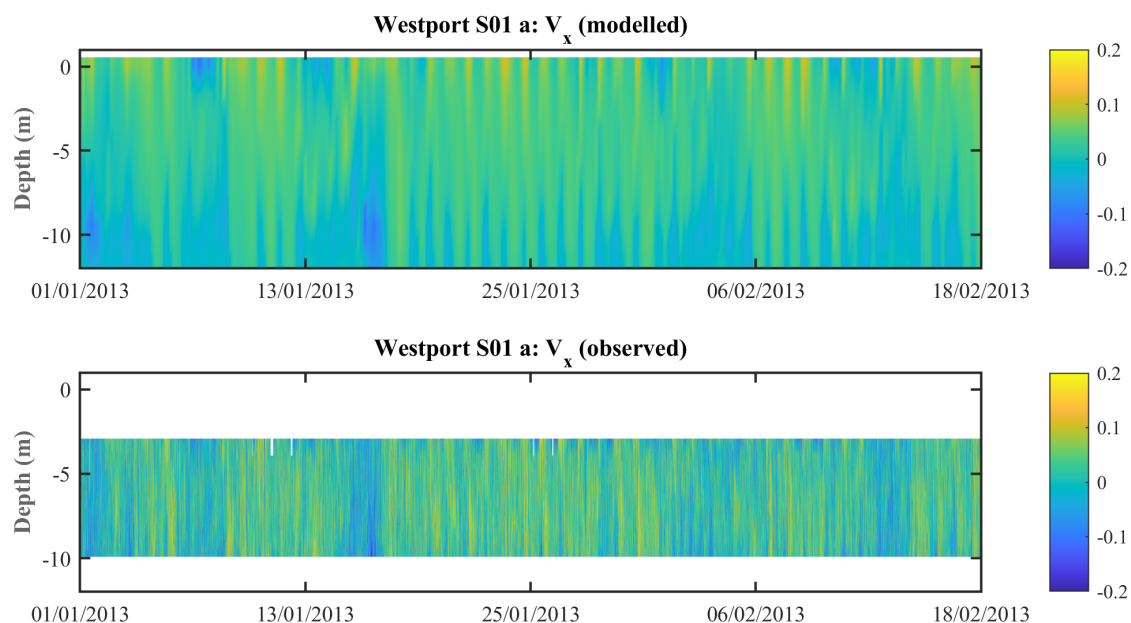


Figure 2.6 Example site profiling output for eastern current speeds at site of S01 of Cockburn Sound.

2.7 sheet map plotting module

The sheet map plotting module can be configured to plot sheet maps or movies of selected water quality variables within define model domain. The module also has options to include m_map tool package for large domain and add current vectors on the map. A summary of sheet map module configuration is provided in Table 2.7. Example configuration and output are provided after the table.

Configuration Summary

Table 2.7 Summary of sheet map module configuration

Field name	Description	Options	Comments
start_plot_ID	select which variable to start plotting		refer to the setting in master.varname
end_plot_ID	select which variable to finish plotting		refer to the setting in master.varname
plotdepth	choose from surface or bottom layer	surface or bottom	
Style	choose from 'm_map' or 'none'	'm_map' or 'none'	
plottype	choose from 'figure' or 'movie'	'figure' or 'movie'	
meshtype	choose from 'patch' or 'meshgrid'	'patch': use original TFV mesh; 'meshgrid': interpolate TFV outputs into mesh grid;	
add_quiver	Option to include current vectors	1: include; 0: not include	
add_coastline	Option to include coastal line onto map	1: include; 0: not include	
coastline_file	File path of the coastline data		Work only when add_coastline = 1; and the data is saved in a .mat format;
xlim	Define the x axis limit		
ylim	Define the y axis limit		
FileFormat	define movie format	choose mp4 or avi	
Quality	define movie quality		100 by default
FrameRate	define movie frame rate		6 by default
resolution	define movie/figure resolution		[1024,768] by default
colormap	define color map type	check matlab help for colormap options	
save_image	option to save frames to images	1: save; 0: not save;	
datearray	define time period for move		
dateformat	define date format in movie		
plot_interval	define interval to plot		refer to model output intervals
outputdirectory	define directory to save plots		
htmloutput	define directory to save HTML files		

clip_depth	define minimum depth to remove NaN values		0.05 by default
istitled	option to add title	1: add; 0: not add	1 by default
isColorbar	option to add colorbar	1: add; 0: not add	1 by default
isylabel	option to add y label	1: add; 0: not add	1 by default
islegend	option to add legend	1: add; 0: not add	1 by default
isYlim	option to define Y axis limits	1: add; 0: not add	0 by default
isGridon	option to add grid on	1: add; 0: not add	1 by default
dimensions	define figure dimensions in centimeters	[20 10] by default	
cAxis.value	define color caxis limits	empty [] by default, matlab will automatically adjust the y limit	

Configuration Example

Example sheet map module configuration

```
sheet.start_plot_ID = 1;
sheet.end_plot_ID = 1;

sheet.style = 'm_map'; % choose 'm_map'(default) or 'none' to use 'm_map14' library
sheet.add_ruler = 1; % 1/0: add/no ruler (default), work only for m_map option
sheet.plotdepth = {'bottom'}; % {'surface','bottom'} Cell-array with either one
sheet.meshstyle = 'meshgrid'; % choose 'patch' or 'meshgrid'
sheet.add_quiver = 1; % 1/0: add/no current vector, work only for 'meshgrid' option
sheet.add_coastline = 1; % 1/0: add/no coast line
sheet.coastline_file = './GIS/Boundary.shp'; % 1/0: add/no coast line

sheet.plottype = 'movie'; % choose 'movie' or 'figure';
sheet.xlim=[115.6666 115.7786]; %[115.6089 115.7786]; %
sheet.ylim=[-32.3000 -32.1333]; %[-32.3206 -31.9921]; %

if strcmpi(sheet.plottype,'movie')
    sheet.FileFormat='mp4'; % choose 'mp4' or 'avi'
    sheet.Quality =100; % movie quality
    sheet.FrameRate =6; % frame rate
    sheet.resolution = [1024,768]; % frame resolution
    sheet.colormap = 'jet'; % colormap options, see Matlab manual
    sheet.save_images = 1; % option to save slide images
    sheet.datearray = [datetime(2021,07,01) datetime(2021,07,21)];
    sheet.dateformat = 'mm/yyyy';
    sheet.plot_interval = 6;
elseif strcmpi(sheet.plottype,'figure')
    sheet.datearray = [datetime(2021,07,01) datetime(2021,07,21)];
    sheet.resolution = [1024,768]; % frame resolution
    sheet.colormap = 'jet'; % colormap options, see Matlab manual
else
    msg=['Error: type of ',sheet.plottype,' is not recognized'];
    error(msg);
end

sheet.outputdirectory = 'plotting/sheet/';

sheet.clip_depth = 0.05; % remove the shallow NaN cells
sheet.cAxis(1).value = [28 35];

sheet.dimc = [0.9 0.9 0.9]; % dimmest (lightest) color
sheet.istitled = 1;
sheet.isColorbar = 1;
sheet.Xlim = [ ];
sheet.Ylim = [ ];
sheet.isAxison = 1;
sheet.dimensions = [20 10]; % Width & Height in cm
```

Example Output

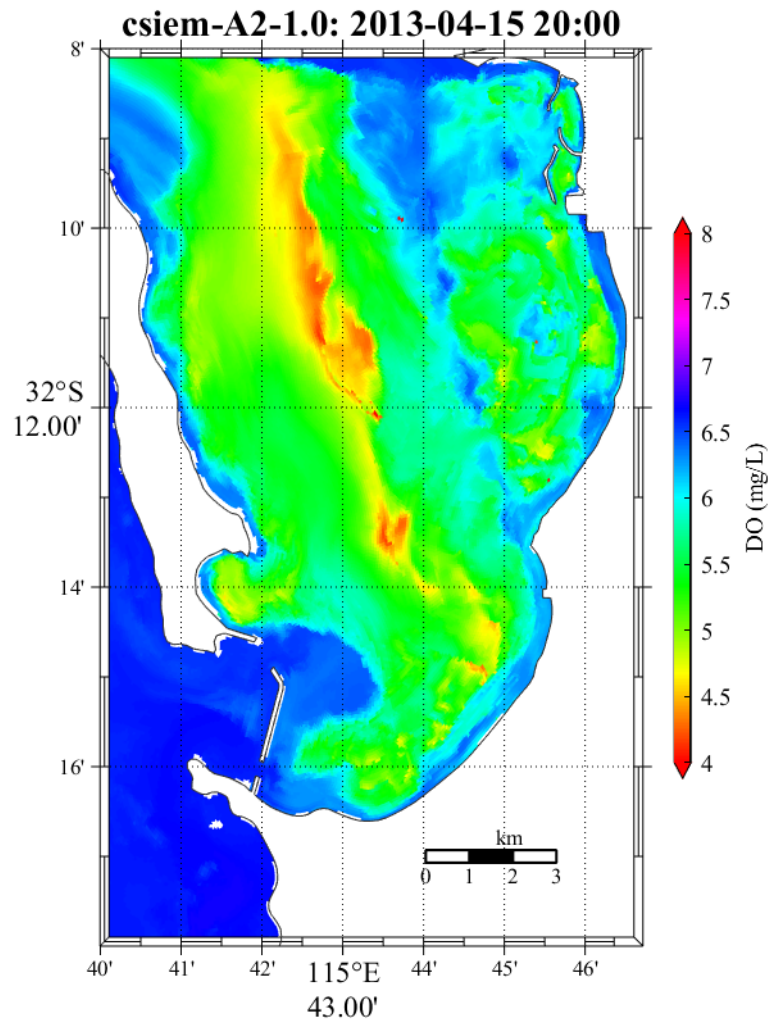


Figure 2.7 Example sheet map output for bottom oxygen at Western Australia coastal area.

2.8 curtain plotting module

The curtain plotting module can also be configured to plot curtain views in pictures or movies of selected water quality variables along a defined polyline. The module also has options to add current vectors on the map. A summary of curtain module configuration is provided in Table 2.8. Example configuration and output are provided after the table.

Configuration Summary

Table 2.8 Summary of curtain plotting module configuration

Field name	Description	Options	Comments
start_plot_ID	select which variable to start plotting		refer to the setting in master.varname
end_plot_ID	select which variable to finish plotting		refer to the setting in master.varname
meshtype	choose from 'patch' or 'meshgrid'	'patch': use original TFV mesh; 'meshgrid': interpolate TFV outputs into mesh grid;	
add_quiver	Option to include current vectors	1: include; 0: not include	
add_coastline	Option to include coastal line onto map	1: include; 0: not include	
polyline	File path of the polyline		
Quality	define movie quality		100 by default
FrameRate	define movie frame rate		6 by default
resolution	define movie/figure resolution		[1024,768] by default
colormap	define color map type	check matlab help for colormap options	
save_image	option to save frames to images	1: save; 0: not save;	
datearray	define time period for move		
dateformat	define date format in movie		
plot_interval	define interval to plot		refer to model output intervals
outputdirectory	define directory to save plots		
htmloutput	define directory to save HTML files		
clip_depth	define minimum depth to remove NaN values		0.05 by default
istitled	option to add title	1: add; 0: not add	1 by default
isColorbar	option to add colorbar	1: add; 0: not add	1 by default
xlim	define limits in X axis		
xticks	define ticks in X axis		
xlabel	define label on X axis		
isylabel	option to add y label	1: add; 0: not add	1 by default

islegend	option to add legend	1: add; 0: not add	1 by default
isYlim	option to define Y axis limits	1: add; 0: not add	0 by default
isGridon	option to add grid on	1: add; 0: not add	1 by default
dimensions	define figure dimensions in centimeters	[20 10] by default	
cAxis.value	define limits of Y axis	empty [] by default, matlab will automatically adjust the y limit	
colorbarposition	define position of color bar		

Configuration Example

Example curtain plotting configuration

```
curtain.start_plot_ID = 1;
curtain.end_plot_ID = 1;
curtain.polyline = 'W:/Jayden/plotting/Curtain_polyline_100m_QC.shp';
curtain.meshstype = 'meshgrid'; % choose 'patch' or 'meshgrid'
curtain.add_quiver = 1; % 1/0: add/no current vector, work only for 'meshgrid' option

curtain.plottype = 'movie'; % choose 'movie' or 'figure';

if strcmpi(curtain.plottype,'movie')
    curtain.FileFormat='mp4'; % choose 'mp4' or 'avi'
    curtain.Quality =100; % movie quality
    curtain.FrameRate =6; % frame rate
    curtain.resolution = [1024,768]; % frame resolution
    curtain.colormap = 'jet'; % colormap options, see Matlab manual
    curtain.save_images = 1; % option to save slide images
    curtain.datearray = [datetime(2021,07,01) datetime(2021,07,21)];
    curtain.dateformat = 'mm/yyyy';
    curtain.plot_interval = 6;
elseif strcmpi(curtain.plottype,'figure')
    curtain.datearray = [datetime(2021,07,01) datetime(2021,07,21)];
    curtain.resolution = [1024,768]; % frame resolution
    curtain.colormap = 'jet'; % colormap options, see Matlab manual
else
    msg=['Error: type of ',curtain.plottype,' is not recognized'];
    error(msg);
end

curtain.outputdirectory = 'plotting/curtain/';

curtain.clip_depth = 0.05; % remove the shallow NaN cells

curtain.dimc = [0.9 0.9 0.9]; % dimmest (lightest) color
curtain.istitled = 1;
curtain.isColorbar = 1;
curtain.xlim = [0 45];% xlim in KM
curtain.xticks = [0:5:45];
curtain.xlabel = 'Distance from CS South (km)';
curtain.isAxison = 1;
curtain.dimensions = [20 10]; % Width & Height in cm
curtain.colorbarposition = [0.92 0.25 0.01 0.45];
curtain.cAxis(1).value = [28 35];
```

Output Example

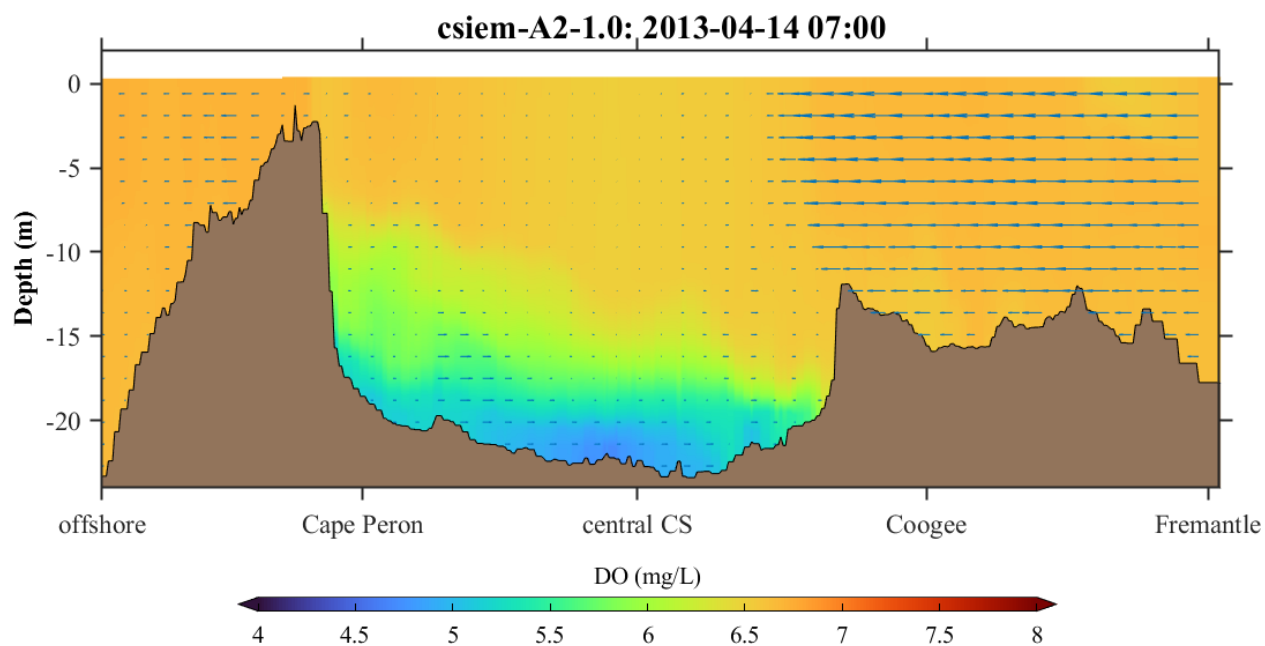


Figure 2.8 Example curtain plot output for dissolved oxygen along a polyline of Cockburn Sound.

References

- Hipsey, M.R., Kilminster, K., Busch, B.D., Bruce, L.C. and Larsen, S. (2014) Modelling oxygen dynamics in the Upper Swan estuary and Canning Weir Pool. *AED Report #R25*, The University of Western Australia, Perth, Australia. 99pp.
- Hipsey, M.R., Kilminster, K., Busch B., 2016a. The Swan-Canning Estuary Response Model (SCERM) v1: Model validation and performance assessment. *AED Report #R29*, The University of Western Australia, Perth, Australia. 42pp.
- Hipsey, M.R., Kilminster, K., Robinson, S., Gedaria, A., Trayler, K., 2016b. The Swan-Canning Estuary Response Model (SCERM) v1: Model Science Basis and Parameterisation. *AED Report #R28*, The University of Western Australia, Perth, Australia. 50pp.
- Bennett, N.D., Croke, B.F.W., Guariso, G., Guillaume, J.H.A., Hamilton, S.H., Jakeman, A.J., Marsili-Libelli, S., Newham, L.T.H., Norton, J.P., Perrin, C., Pierce, S. A., Robson, B., Seppelt, R., Voinov, A.A., Fath, B.D., Andreassian, V., 2013. Characterising performance of environmental models. *Environ. Model. Software* 40, 1–20.
- Hipsey, M.R. et al., 2020. A system of metrics for the assessment and improvement of aquatic ecosystem models. *Environmental Modelling and Software*, 128, 104697.